

URZHAR, Kazakhstan

WMO: 366390

Lat: 47.117N

Lon: 81.617E

Elev: 491

StdP: 95.56

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.7	-29.5	-34.8	0.1	-32.6	-31.9	0.2	-29.3	7.8	-9.0	5.4	-7.7	0.3	130	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	34.7	18.7	33.0	18.1	31.3	17.7	20.2	30.0	19.5	29.1	18.8	28.3	1.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	13.2	22.9	16.4	12.4	22.5	15.5	11.7	22.0	60.5	30.1	57.8	29.2	55.4	28.3	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.4	6.3	4.7	-35.8	37.5	3.3	1.7	-38.2	38.7	-40.1	39.7	-42.0	40.6	-44.4	41.9		
			-35.7	22.2	3.3	1.5	-38.1	23.2	-40.0	24.1	-41.9	24.9	-44.3	26.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-14.6	-11.4	-2.8	8.9	14.9	20.1	22.0	20.4	14.6	7.3	-1.7	-11.2	(6)
(7)		DBStd	13.83	6.44	6.05	6.27	5.10	3.84	3.23	2.81	3.32	4.28	4.56	6.49	6.78	(7)
(8)		HDD10.0	2973	763	600	397	76	6	0	0	11	108	352	659		(8)
(9)		HDD18.3	4936	1022	833	655	282	117	19	3	17	124	343	602	917	(9)
(10)		CDD10.0	1372	0	0	1	44	159	302	371	322	148	23	1	0	(10)
(11)		CDD18.3	292	0	0	0	1	12	71	116	81	12	0	0	0	(11)
(12)		CDH23.3	4977	0	0	0	46	296	1087	1657	1470	406	15	0	0	(12)
(13)		CDH26.7	2108	0	0	0	8	79	441	766	682	131	1	0	0	(13)
(14)	Wind	WSAvg	1.1	0.5	0.6	1.0	1.9	1.8	1.2	1.0	1.2	1.0	0.7	0.6	(14)	
(15)	Precipitation	PrecAvg	443	43	34	27	38	37	29	36	18	20	52	61	49	(15)
(16)		PrecMax	690	131	84	85	94	109	92	129	51	62	161	143	116	(16)
(17)		PrecMin	245	3	2	2	1	1	2	1	1	0	8	7	5	(17)
(18)		PrecStd	118	28	20	21	30	31	21	31	14	14	37	38	32	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.8	3.5	17.9	27.4	31.7	34.9	37.3	36.6	33.5	25.3	16.3	3.6	(19)
(20)			MCWB	-0.5	0.2	9.3	15.1	16.9	19.3	20.2	19.0	17.7	13.7	8.8	0.5	(20)
(21)		2%	DB	-0.9	1.3	12.7	24.3	28.9	32.8	34.9	34.6	30.1	22.5	12.6	1.0	(21)
(22)			MCWB	-1.9	-0.8	7.1	13.4	15.9	18.3	19.0	18.3	16.2	12.2	6.7	-0.5	(22)
(23)		5%	DB	-2.8	-0.2	8.6	21.8	26.6	31.2	32.9	32.8	27.9	19.8	9.2	-0.5	(23)
(24)			MCWB	-4.0	-1.8	4.7	12.1	14.9	17.9	18.4	17.8	15.5	11.0	5.4	-1.8	(24)
(25)		10%	DB	-4.7	-2.0	5.5	19.4	24.5	29.4	31.0	30.7	25.5	16.8	6.3	-2.0	(25)
(26)			MCWB	-5.6	-3.5	2.3	10.9	14.3	17.4	18.1	17.3	14.4	9.7	3.8	-3.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.1	0.8	10.2	16.1	18.4	21.1	21.7	20.7	18.4	14.3	9.9	1.1	(27)
(28)			MCD B	0.7	2.4	16.6	25.1	28.4	30.5	32.0	29.8	30.6	23.6	14.2	2.6	(28)
(29)		2%	WB	-1.8	-0.4	7.5	14.1	17.0	19.9	20.5	19.5	17.1	12.8	7.4	-0.3	(29)
(30)			MCD B	-1.0	1.0	12.4	22.7	25.7	29.4	30.5	30.1	27.9	21.1	11.6	0.8	(30)
(31)		5%	WB	-3.9	-1.8	4.9	12.7	16.1	19.1	19.6	18.8	15.9	11.5	5.6	-1.7	(31)
(32)			MCD B	-3.0	-0.3	8.4	20.1	23.8	27.8	29.1	29.6	26.1	18.5	8.7	-0.7	(32)
(33)		10%	WB	-5.6	-3.5	2.6	11.5	15.1	18.2	18.9	18.1	14.9	10.1	3.9	-3.1	(33)
(34)			MCD B	-4.6	-2.1	5.3	18.1	22.6	26.9	28.1	28.7	24.2	16.1	6.3	-2.1	(34)
(35)	Mean Daily Temperature Range	MDBR	13.0	14.5	12.5	14.0	14.9	15.3	15.7	17.7	17.8	15.0	12.0	12.3	(35)	
(36)		5% DB	MCD BR	11.8	14.3	14.3	19.7	18.5	18.5	19.2	21.0	21.5	20.6	16.0	11.7	(36)
(37)			MCWBR	10.5	12.0	9.2	10.2	8.2	7.2	6.9	7.9	9.9	11.4	10.2	9.7	(37)
(38)		5% WB	MCD BR	10.6	13.1	13.8	17.1	15.7	15.9	16.2	17.8	20.0	18.5	14.6	10.1	(38)
(39)			MCWBR	9.6	11.0	8.9	9.4	7.9	7.2	6.6	7.5	9.7	10.6	9.6	8.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.313	0.346	0.398	0.446	0.431	0.434	0.442	0.402	0.369	0.370	0.349	0.311	(40)	
(41)		taud	2.029	1.992	1.990	1.995	2.097	2.158	2.156	2.250	2.285	2.175	2.101	2.052	(41)	
(42)		Ebn at Noon	725	785	804	804	835	833	820	840	834	756	678	674	(42)	
(43)		Edh at Noon	98	127	150	164	154	146	144	126	111	105	92	86	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.79	2.82	3.98	5.23	6.64	6.94	6.61	6.11	4.78	2.90	1.63	1.32	(44)	
(45)		RadStd	0.11	0.14	0.28	0.35	0.29	0.37	0.40	0.31	0.28	0.25	0.15	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	-0.40	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page